

Demo to production, without rewriting from scratch.

Nine probes of empirical evidence that **extracting the spec** from a working demo is the hard part — and it's tractable.

A great demo is the worst brief.

- The prototype **works** — and customers want it.
- Every quirk, layout choice, and business rule is already decided in code.
- Nobody wrote the spec. The code **is** the spec.
- Hand-rewrites lose intent and cost months.



MONTHS OF ENG TIME, LOST DESIGN INTENT, AND A
REWRITE THAT DOESN'T QUITE MATCH THE DEMO THAT WON
THE DEAL.

You can't write the spec by hand — the demo already **is** one.

The prototype encodes hundreds of decisions: status-pill colors, filter semantics, empty-state copy, the exact 8-column table schema, the waterfall math on the balance cards.

A human spec captures maybe 10%. The rest shows up as "bug reports" months later — when it's the **rewrite** that's wrong, not the demo.

Extracting the spec is hard. Rebuilding from a good spec is **tractable**.

Conventional wisdom: "code generation is the hard part."

We invert this. Modern LLMs can rebuild competently from a clean spec — if you give them one. The work is turning the messy demo **into** that spec.

Two stages.

1 · Reverse-spec-extraction

Four probe agents walk the demo in parallel:

- `ui-walker` — clicks every screen
- `static` — reads the code
- `data` — pulls sample rows & PDFs
- `external` — maps API calls

A reconciler merges them into an **artifact pack** — 6 structured files + screenshots.

2 · Ralph-loop build

A build-controller slices the work by **user journey** and dispatches per-slice executors.

Each slice is verified by:

- oracle test suite
- Playwright journey walks
- **side-by-side screenshot check** against the prototype

Nine probes. Convergence.

PROBE	WHAT IT TESTED	OUTCOME
0 – 1.5	Ingest pipeline	Spec extraction works end-to-end
2	First full build from spec	4.5K lines, 45 endpoints, 11 pages — zero human help
2.5	Stronger oracle + vertical slices	API layer excellent; UI still a shell
2.6	Playwright journey tests	Orphan pages in 6 of 11 screens
2.7	Slice-by-journey	Orphans fixed; dialog theater surfaces
2.8	Seeded data + screenshot-as-contract	Live app renders real data, 1:1 with reference
2.9	Author/judge separation	In flight

Each probe isolates one layer, finds its failure mode, and fixes it. Discipline, not luck.

THE MOMENT OF PROOF

Same structure.
Real data.

Calibrax Grid

FINANCE LEDGER

CLIENT PROBE

16 Apr. 2026 12:55:08

Activity

DESIGN

Business Context

Workflow Design

DEPLOY

The Stage

MANAGE

Accounts Receivable

Accounts Payable

Bank Reconciliation

General Ledger

Analytics

Activity

Dashboard

CONFIGURE

Minion Library

Reset Data

Accounts Receivable

FY2026 · Trade Receivables · Account 1100 · Active period: FY2026

Export CSV

Upload Documents

OPENING BALANCE

US\$ 0.00

1 Jan 2026 brought forward

INVOICED

+ US\$ 0.00

COLLECTED

- US\$ 0.00

CLOSING BALANCE

US\$ 0.00

17 Apr 2026 current

TOTAL INVOICED

US\$ 0

0 invoices · FY2026

COLLECTED

US\$ 0

0 paid · 0 pending match

OUTSTANDING

US\$ 0

excl. pending match

OVERDUE / DISPUTED

US\$ 0

None

FILTER:

All Statuses

All Sources

Counterparty

☐ Hide Shopify

☐ Hide Aged

0 of 0 active entries

Chart of Accounts not configured

Set up a Chart of Accounts before uploading AR/AP documents so transactions can be assigned to the correct account codes. [Go to Business Context](#)

SERIES	DESCRIPTION	DATE	DUE DATE	AMOUNT	ACCT CODE	STATUS	ACTIONS
No AR data for PROBE							

REFERENCE · prototype AR page on empty tenant (all US\$ 0)

07 · PROOF

Calibrax Grid

FINANCE LEDGER

CLIENT PROBE

17 Apr. 2026 14:04:50

Activity

Accounts Receivable

FY2026 · Trade Receivables · Account 1100 · Active period: FY2026

Export CSV

Upload Documents

OPENING BALANCE

US\$ 0.00

1 Jan 2026 brought forward

INVOICED

+ US\$ 254,796.36

COLLECTED

- US\$ 35,408.97

CLOSING BALANCE

US\$ 219,387.39

17 Apr 2026 current

TOTAL INVOICED

US\$ 254,796

37 invoices · FY2026

COLLECTED

US\$ 35,409

4 paid · 23 pending match

OUTSTANDING

US\$ 170,256

excl. pending match

OVERDUE / DISPUTED

US\$ 49,131

6 flagged

FILTER:

All Statuses

All Sources

All counterparties

☐ Hide Shopify

☐ Hide Aged

37 of 37 active entries

Chart of Accounts not configured

Set up a Chart of Accounts before uploading AR/AP documents so transactions can be assigned to the correct account codes. [Go to Business Context](#)

SERIES	DESCRIPTION	DATE	DUE DATE	AMOUNT	ACCT CODE	STATUS	ACTIONS
AR-010	Katering Guru-Guru Maktab INVOICE · INV/2026/010	06 Nov 2025	-	US\$ 4,741.80	-	OVERDUE	...
AR-010	Katering Guru-Guru Maktab DELIVERY_ORDER · DO/2026/010	08 Nov 2025	-	US\$ 4,741.80	-	OVERDUE	...
AR-010	Katering Guru-Guru Maktab GRN · GRN/2026/010	09 Nov 2025	-	US\$ 4,741.80	-	OVERDUE	...
AR-003	Giant Hypermarket (Kepong) INVOICE · INV/2026/003	31 Jan 2026	-	US\$ 19,158.02	-	PENDING MATCH	...
AR-003	Giant Hypermarket (Kepong) DELIVERY_ORDER · DO/2026/003	02 Feb 2026	-	US\$ 19,158.02	-	PENDING MATCH	...
AR-002	Syarikat Katering Harmoni Sdn. Bhd. INVOICE · INV/2026/002	03 Feb 2026	-	US\$ 6,109.42	-	PENDING MATCH	...
AR-003	Giant Hypermarket (Kepong) GRN · GRN/2026/003	03 Feb 2026	-	US\$ 19,158.02	-	DISPUTED	...
AR-002	Syarikat Katering Harmoni Sdn. Bhd. DELIVERY_ORDER · DO/2026/002	04 Feb 2026	-	US\$ 6,109.42	-	PENDING MATCH	...
AR-008	Koperasi Kakitangan Hospital (Pantai) INVOICE · INV/2026/008	04 Feb 2026	-	US\$ 3,262.68	-	PENDING MATCH	...
AR-002	Syarikat Katering Harmoni Sdn. Bhd. GRN · GRN/2026/002	05 Feb 2026	-	US\$ 6,109.42	-	PENDING MATCH	...
AR-008	Koperasi Kakitangan Hospital (Pantai) DELIVERY_ORDER · DO/2026/008	05 Feb 2026	-	US\$ 3,262.68	-	PENDING MATCH	...
AR-008	Koperasi Kakitangan Hospital (Pantai) GRN · GRN/2026/008	06 Feb 2026	-	US\$ 3,262.68	-	PENDING MATCH	...
AR-006	Restoran Nelayan Seafood INVOICE · INV/2026/006	07 Feb 2026	-	US\$ 1,950.93	-	PENDING MATCH	...
AR-007	Kedai Makanan Muslim Al-Amin INVOICE · INV/2026/007	07 Feb 2026	-	US\$ 2,471.18	-	PENDING MATCH	...
AR-005	Sekolah Menengah Kebangsaan Damansara Canteen INVOICE · INV/2026/005	08 Feb 2026	-	US\$ 799.66	-	PENDING MATCH	...
AR-006	Restoran Nelayan Seafood DELIVERY_ORDER · DO/2026/006	08 Feb 2026	-	US\$ 1,950.93	-	PENDING MATCH	...
AR-007	Kedai Makanan Muslim Al-Amin DELIVERY_ORDER · DO/2026/007	08 Feb 2026	-	US\$ 2,471.18	-	PENDING MATCH	...

We learn the system by breaking it.

Orphan pages (2.6)

Slicing by database table left the cross-cutting pages (Dashboard, Analytics) as scaffolds. **Fix:** slice by user journey instead.

Dialog theater (2.7)

Buttons opened dialogs that closed without doing anything. **Fix:** every task must assert **post-action state**, not just "dialog appears."

Empty-tenant blind spot (2.7)

The build never rendered a populated row. **Fix:** seed the prototype's own data before journey walks.

Self-evaluation (2.8)

When the same agent plans the work **and** judges its own plan, it passes every gate while cutting scope. **Fix:** separate author from judge (probe 2.9, in flight).

A fast path from prototype to product.

Keep momentum

Prototype in days, productize in days. Design intent survives — because the prototype **is** the spec.

Repeatable

The method works per-probe, so it works per-prototype. Each new Calibrax demo plugs into the same pipeline.

Disciplined

Every failure becomes a probe, a fix, and a piece of institutional memory — not a regression.

The prototype isn't a deliverable anymore. It's **upstream of the product.**

From rewrite tool to **rewrite-and-harden** tool.

NOW

Probe 2.9 · plan-reviewer

A dedicated fresh-context reviewer, armed with a calibration set of past mistakes.

Tests whether same-model (Claude-on-Claude) separation catches the self-

NEAR-TERM · TWO TRACKS

App shell scaffold

Pre-built monorepo: shared components, schema helpers, seed bootstrap, locked file layout. Probes start at the slice-loop, not from empty.

Multi-model reviewer

Codex alongside **plan-**

BIGGER SHIFT

Constrained artifact improvement loop

Between extract and build, insert specialized hardeners:

data, **api-envelope**, **a11y**, **missing-index**, **error-boundary**.

v1 — constrained: same behavior as demo, but hardened. v2 — **strictly better than the demo**.

What would help.

- **A second prototype to probe against.**
Pick one from the team — ideally with a real customer story behind it.
- **30 minutes with a PM** who's lived a demo → production rewrite. Where did **your** spec leak?
- **Honest pushback on the thesis.** If "extracting the spec is the hard part" is wrong, the sooner we know, the better.

When the tool learns the system by breaking it, each failure is evidence — not a setback.

— OPERATING PRINCIPLE

THANK YOU

Questions, objections, prototypes.

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